ICS 2305 : SYSTEMS PROGRAMMING
GROUP ASSIGNMENT

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Question 1

You are tasked with building a dynamic patient records system for a hospital using C. The system must:

- Add new patients (allocate memory for patient details: name, ID, medical history)
- Update records (expand/shrink memory if medical history updates)
- Remove discharged patients (free memory)
- Handle errors (e.g., failed allocations, NULL checks)

Implementation

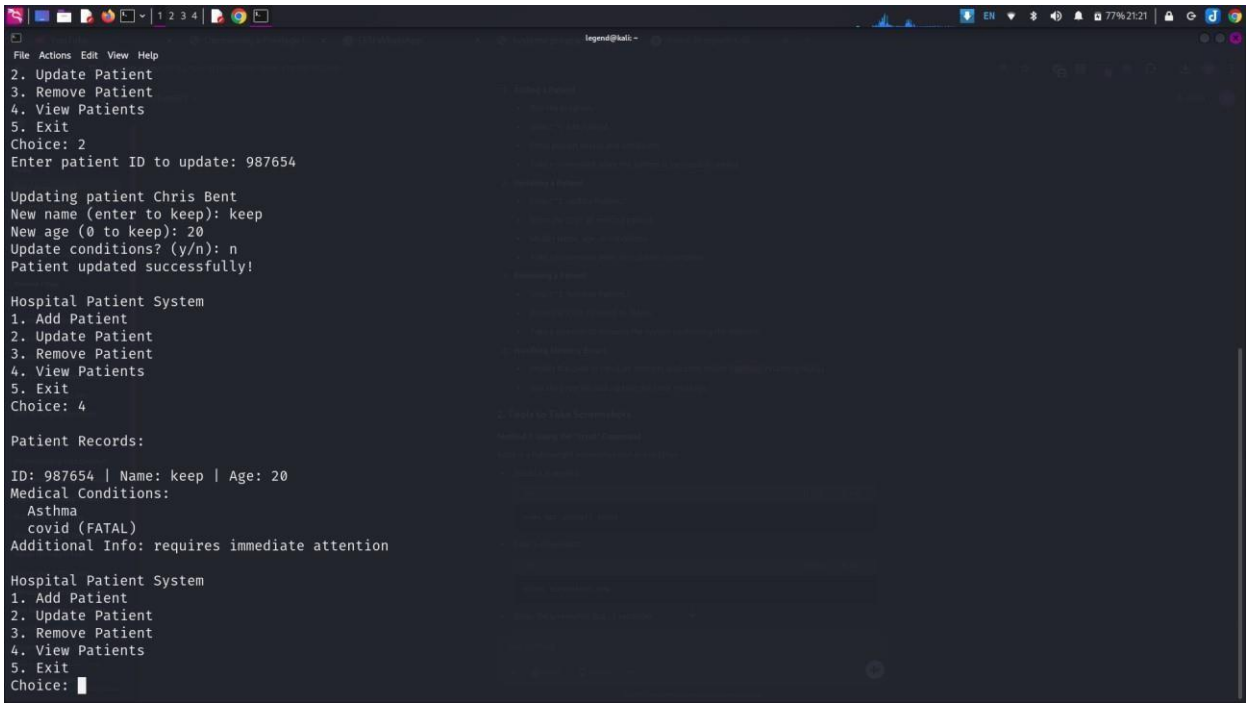
Code attached

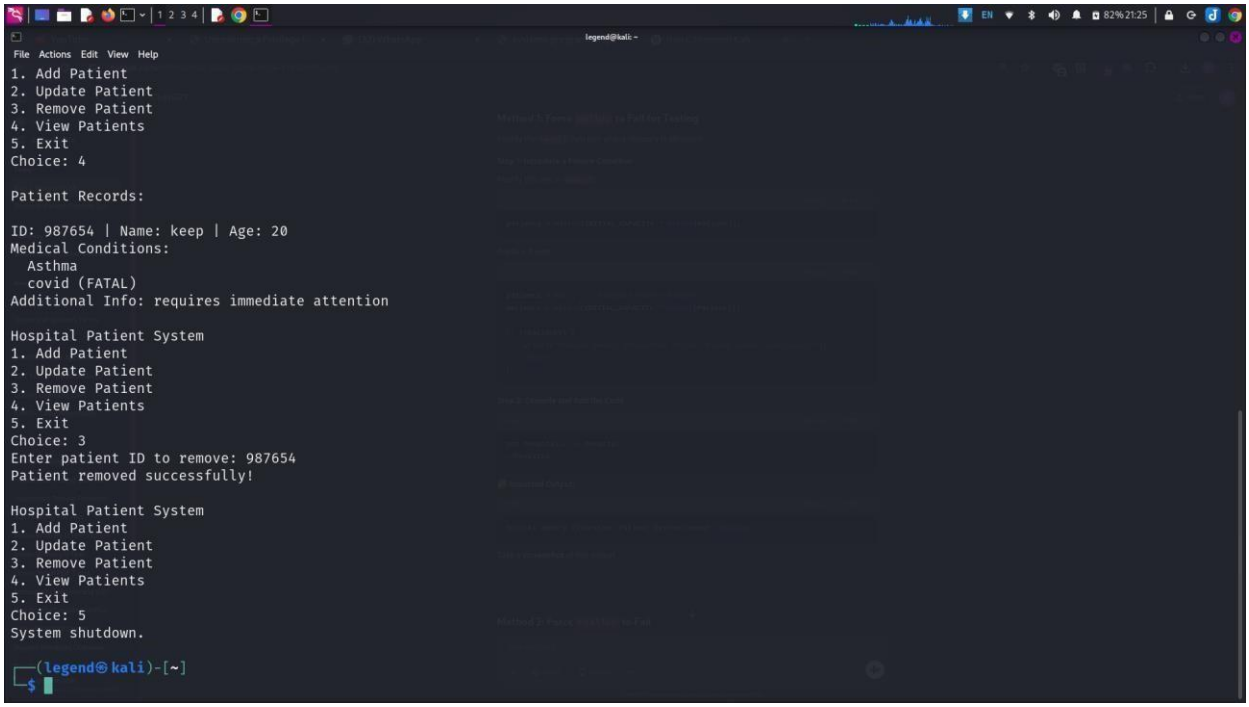
Output




File Actions Edit View Help
(legend㉿kali)-[~]
$ gcc patient_system.c -o patient
(legend㉿kali)-[~]
$ ./patient
Hospital Patient System
1. Add Patient
2. Update Patient
3. Remove Patient
4. View Patients
5. Exit
Choice: 1
Enter patient ID: 987654
Enter name: Chris Bent
Enter age: 19
Condition name (or 'none'): Asthma
Fatal? (y/n): n
Add another? (y/n): y
Condition name (or 'none'): covid
Fatal? (y/n): y
Add another? (y/n): n
Other info: requires immediate attention
Patient added successfully!
Hospital Patient System
1. Add Patient
2. Update Patient
3. Remove Patient
4. View Patients
5. Exit
Choice:


File Actions Edit View Help
Choice: 1
Enter patient ID: 987654
Enter name: Chris Bent
Enter age: 19
Condition name (or 'none'): Asthma
Fatal? (y/n): n
Add another? (y/n): y
Condition name (or 'none'): covid
Fatal? (y/n): y
Add another? (y/n): n
Other info: requires immediate attention
Patient added successfully!
Hospital Patient System
1. Add Patient
2. Update Patient
3. Remove Patient
4. View Patients
5. Exit
Choice: 2
Enter patient ID to update: 987654
Updating patient Chris Bent
New name (enter to keep): keep
New age (0 to keep): 20
Update conditions? (y/n): n
Patient updated successfully!
Hospital Patient System
1. Add Patient
2. Update Patient
3. Remove Patient
4. View Patients
5. Exit
Choice:


File Actions Edit View Help
2. Update Patient
3. Remove Patient
4. View Patients
5. Exit
Choice: 2
Enter patient ID to update: 987654
Updating patient Chris Bent
New name (enter to keep): keep
New age (0 to keep): 20
Update conditions? (y/n): n
Patient updated successfully!
Hospital Patient System
1. Add Patient
2. Update Patient
3. Remove Patient
4. View Patients
5. Exit
Choice: 4
Patient Records:
ID: 987654 | Name: keep | Age: 20
Medical Conditions:
Asthma
covid (FATAL)
Additional Info: requires immediate attention
Hospital Patient System
1. Add Patient
2. Update Patient
3. Remove Patient
4. View Patients
5. Exit
Choice:


File Actions Edit View Help
1. Add Patient
2. Update Patient
3. Remove Patient
4. View Patients
5. Exit
Choice: 4
Patient Records:
ID: 987654 | Name: keep | Age: 20
Medical Conditions:
Asthma
covid (FATAL)
Additional Info: requires immediate attention
Hospital Patient System
1. Add Patient
2. Update Patient
3. Remove Patient
4. View Patients
5. Exit
Choice: 3
Enter patient ID to remove: 987654
Patient removed successfully!
Hospital Patient System
1. Add Patient
2. Update Patient
3. Remove Patient
4. View Patients
5. Exit
Choice: 5
System shutdown.
(legend㉿kali)-[~]
$

Question 2

Design a C-based system to manage traffic light data for a smart city. The system must:

- Dynamically allocate/deallocate memory for traffic sensors
- Handle buffer overflows (e.g., corrupt sensor data)
- Prevent memory leaks (e.g., faulty sensor removal)
- Simulate garbage collection for orphaned data

Implementation

Code attached

Sample Output

